Number 3 — Prime factors · HCF and LCM · Ratio

No calculator. Marks in brackets. ★ = harder question.

1 Express 1260 as a product of its prime factors, in index form. [4]

2 Find the HCF and the LCM of 48 and 180. [4]

3 Two teachers set regular tests. Ms Avery's tests always have 20 questions and Mr Baig's always have 36. In one term both classes answer the same total number of questions. [4]

a What is the smallest total number of questions that makes this possible?

b How many tests does Ms Avery's class take in that case?

4 Nadia has 180 pens and 300 notebooks to pack into identical goody bags, using everything. What is the greatest number of bags she can make, and what goes in each bag? [4]

5 Three cousins aged 45, 18 and 9 share €720 in the ratio of their ages. How much does each receive? [4]

6 Theo finds that 45 ml of cordial mixed with 270 ml of water makes a drink he likes. [5]

a Write the ratio cordial : water in its simplest form.

b Theo uses 6 litres of water to make the same strength of drink. What total volume of drink does he make?

7 Which ratio is larger, $5 : 3$ or $17 : 10$? Show how you decide. [2]

8 Two gear wheels have 24 and 40 teeth and are meshed together. A mark is painted where they touch. What is the smallest number of turns of the 24-tooth wheel before the two marks meet again? [3]

Total: 30 marks

Check: "greatest number of equal groups" → HCF · "next time they coincide" → LCM · simplify a ratio before sharing · the HCF × LCM of two numbers equals their product.